

YEKATERINBURG KOLTSOVO, Russian Federation

WMO: 284400

Lat: 56.743N

Lon: 60.803E

Elev: 764

StdP: 14.29

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.8	-18.0	-30.1	1.0	-23.5	-23.8	1.5	-17.4	21.0	15.5	19.0	15.5	4.4	0	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	85.8	67.5	82.3	65.8	78.9	64.3	70.3	80.5	68.3	78.4	66.5	76.0	7.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	100.9	74.2	64.7	94.2	72.3	62.8	88.1	70.4	34.8	80.2	33.1	78.5	31.5	75.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.2	16.2	-30.8	90.8	7.3	3.5	-36.1	93.4	-40.3	95.4	-44.4	97.4	-49.7	100.0
			-31.0	72.8	7.2	2.4	-36.2	74.5	-40.5	75.9	-44.5	77.2	-49.8	78.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	6.5	9.6	23.7	39.2	52.2	61.1	64.6	60.5	50.0	37.8	21.7	11.0	(6)
(7)		DBStd	22.86	12.82	12.55	9.17	9.44	9.10	8.00	6.68	7.51	7.98	8.75	12.90	13.09	(7)
(8)		HDD50	6280	1350	1130	815	340	87	10	1	6	96	387	850	1208	(8)
(9)		HDD65	10532	1815	1550	1280	773	406	164	92	183	453	843	1300	1673	(9)
(10)		CDD50	1405	0	0	0	17	154	344	452	331	97	10	0	0	(10)
(11)		CDD65	181	0	0	0	0	8	48	78	43	4	0	0	0	(11)
(12)		CDH74	2079	0	0	0	7	176	523	837	484	52	0	0	0	(12)
(13)		CDH80	553	0	0	0	1	25	132	231	155	9	0	0	0	(13)
(14)	Wind	WSAvg	7.3	6.8	6.9	8.1	8.0	7.5	7.3	6.2	6.3	6.8	7.9	7.8	7.4	(14)
(15)	Precipitation	PrecAvg	21.6	1.1	0.7	1.0	1.3	2.0	3.0	3.6	2.8	1.9	1.7	1.4	1.1	(15)
(16)		PrecMax	28.0	3.0	1.5	2.4	3.6	4.0	5.4	7.7	5.9	3.6	3.7	3.1	2.4	(16)
(17)		PrecMin	14.7	0.2	0.1	0.2	0.1	0.3	1.1	1.5	0.9	0.3	0.2	0.2	0.3	(17)
(18)		PrecStd	3.5	0.6	0.4	0.6	0.9	0.9	1.2	1.8	1.3	0.8	0.9	0.7	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	37.6	50.0	73.3	82.8	88.1	89.9	89.9	80.4	66.1	47.9	36.3	(19)
(20)			MCWB	32.8	33.9	40.9	54.3	60.4	67.2	68.0	68.2	60.8	53.5	43.7	34.3	(20)
(21)		2%	DB	31.1	34.1	43.2	66.0	79.0	84.0	85.8	84.3	73.6	60.1	42.8	33.9	(21)
(22)			MCWB	29.1	30.9	36.7	50.1	58.9	66.2	68.0	67.5	58.8	49.9	40.9	31.7	(22)
(23)		5%	DB	28.2	31.0	39.2	60.3	74.9	80.2	82.3	79.1	67.9	55.2	39.2	31.8	(23)
(24)			MCWB	26.5	28.3	34.1	47.3	57.7	64.8	67.4	66.1	57.2	47.6	37.0	29.8	(24)
(25)		10%	DB	24.8	28.2	36.2	55.2	70.0	76.3	78.7	74.8	63.0	50.3	36.0	28.6	(25)
(26)			MCWB	22.9	25.7	32.2	45.0	55.0	63.2	65.6	64.4	54.9	45.6	33.9	26.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	34.7	41.3	55.3	65.2	72.8	72.6	72.3	64.2	54.7	44.3	34.7	(27)
(28)			MDCB	35.0	37.5	47.5	70.2	75.3	80.9	81.4	81.2	72.7	63.0	46.7	36.1	(28)
(29)		2%	WB	29.1	31.1	37.2	51.5	61.3	68.9	70.5	69.8	60.8	50.9	40.6	32.0	(29)
(30)			MDCB	30.8	33.7	42.0	61.5	73.2	79.2	80.7	80.3	69.6	57.6	42.1	33.4	(30)
(31)		5%	WB	26.3	28.5	34.7	48.6	58.8	66.6	68.6	67.4	58.2	48.4	37.0	29.8	(31)
(32)			MDCB	27.9	30.9	38.1	58.4	71.4	76.6	78.8	76.9	65.8	54.0	38.7	31.5	(32)
(33)		10%	WB	23.1	25.5	32.6	45.6	56.5	64.4	67.0	65.0	55.9	45.8	34.1	26.7	(33)
(34)			MDCB	24.8	27.9	35.9	54.2	68.0	73.9	76.6	73.3	62.0	50.1	35.6	28.3	(34)
(35)	Mean Daily Temperature Range	MDBR	14.9	18.3	17.9	19.0	22.1	19.9	19.6	18.6	16.9	12.3	11.4	13.2	(35)	
(36)		5% DB	MCDBR	11.5	17.6	20.4	27.9	29.9	26.2	26.8	27.0	24.9	22.0	12.7	10.5	(36)
(37)			MCWBR	10.8	15.0	15.0	15.7	14.7	12.7	12.2	13.1	13.3	13.8	10.9	9.8	(37)
(38)		5% WB	MCDBR	11.6	15.3	16.7	25.2	26.1	23.4	23.4	24.1	22.2	19.7	10.9	10.3	(38)
(39)			MCWBR	11.1	13.5	12.9	15.3	14.7	13.0	12.1	13.2	13.8	13.2	10.1	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.281	0.312	0.365	0.410	0.409	0.404	0.435	0.445	0.383	0.350	0.307	0.278	(40)	
(41)		taud	2.066	2.060	1.973	2.096	2.232	2.317	2.222	2.166	2.308	2.263	2.145	2.025	(41)	
(42)		Ebn at Noon	181	226	242	252	262	266	253	240	240	213	170	143	(42)	
(43)		Edh at Noon	19	29	42	43	40	37	41	40	30	24	18	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	220	531	980	1299	1590	1728	1680	1286	820	403	191	129	(44)	
(45)		RadStd	23	44	75	156	166	155	133	131	62	78	23	21	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) +1.28	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page